

Discussion of Angela Ma

“The Real Effects of Liquidity Relief: Commercial Rent Forbearance in the Covid-19 Recession”

Benedict Guttman-Kenney
Rice University

Midwest Finance Association, 13th March 2026



RICE | BUSINESS

JONES GRADUATE SCHOOL OF BUSINESS
VIRANI UNDERGRADUATE SCHOOL OF BUSINESS



The *first* rule of 2020 is...
You do not talk about 2020



**The *first* rule of 2020 is...
You do not talk about 2020**

**The *second* rule of 2020 is...
You do not talk about 2020**



The *first* rule of 2020 is... The *second* rule of 2020 is...
You do not talk about 2020 You do not talk about 2020

What am I going to do?

Talk about 2020



The *first* rule of 2020 is... The *second* rule of 2020 is...
You do not talk about 2020 You do not talk about 2020

What am I going to do?

Talk about 2020

Why?

2020 is a seismic economic event to learn from!

My Household Finance Perspective On A Corporate Finance Topic

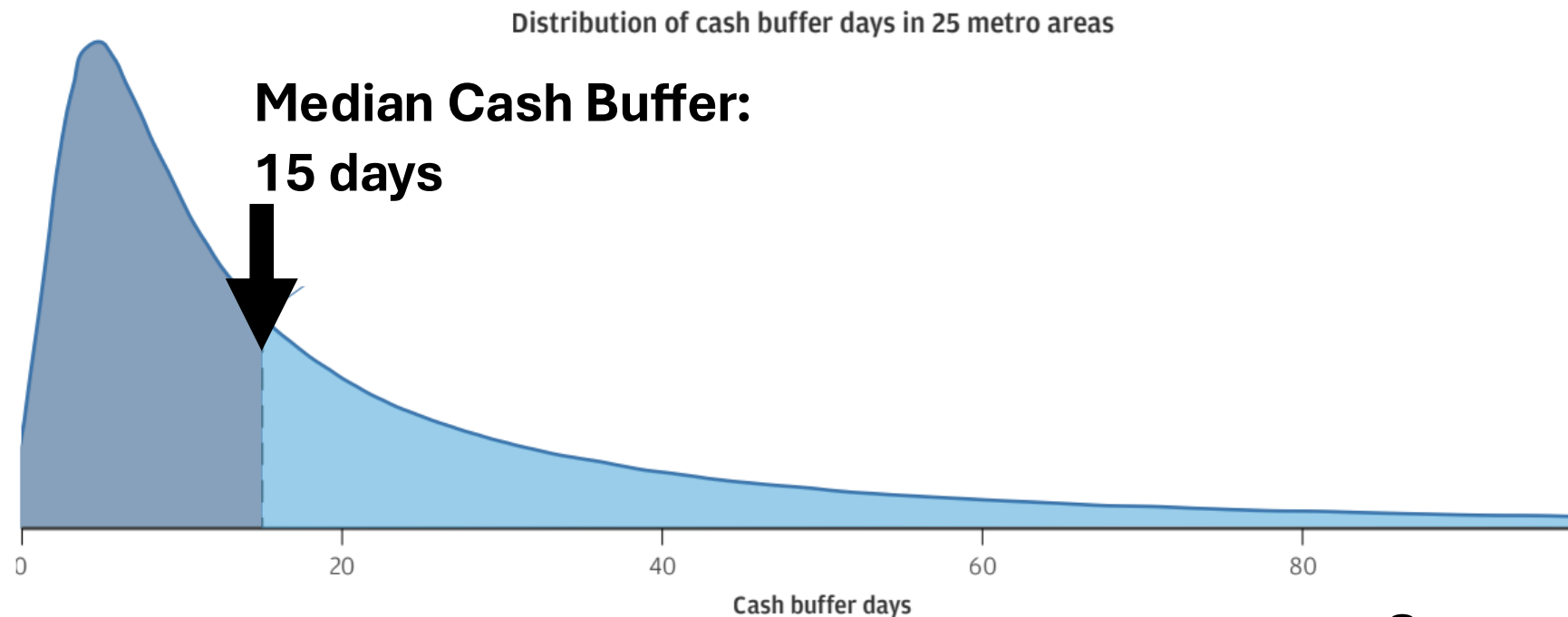
- **Households** often have volatile incomes & limited liquid cash
 - ➔ Vulnerable to income/expenditure shocks
Morduch & Schneider (2017); Ganong et al. (2025); Fulford & Low (2025)
- Growing recognition that **liquidity** drives defaults
Ganong & Noel (2020 AER; 2023 QJE), Low (2025 RFS R&R; 2026 RESTAT)

My Household Finance Perspective On A Corporate Finance Topic

- **Households** often have volatile incomes & limited liquid cash
 - ➔ Vulnerable to income/expenditure shocks
Morduch & Schneider (2017); Ganong et al. (2025); Fulford & Low (2025)
- Growing recognition that **liquidity** drives defaults
Ganong & Noel (2020 AER; 2023 QJE), Low (2025 RFS R&R; 2026 RESTAT)
- Many studies on **what policies help recovery** (e.g., debt or payment relief, stimulus)
Fiorin, Hall, & Kanz (2023 RFS R&R), Aydin (2024 JFE R&R), Catherine & Yannelis (2022 JFE), Di Maggio, Kalda, & Yao (2026 JF), Dobbie & Song (2020 AER) **COVID-19** Cherry, Jiang, Matvos, Piskorski, & Seru (2021 BPEA); Katz (2024); Dinerstein, Yannelis, & Chen (2024 AER: I); Hamdi, Kalda, & Sovich (2025 MS)
- **This paper may benefit linking with literature on household eviction moratoriums**
 - ↓ infections & deaths (Jowers, Timmins, Bhavsar, Hu, & Marshall, 2021)
 - ↑ spending, especially on food (An, Gabriel, & Tzur-Ilan, 2022 AEA P&P)
 - + more broadly (Collinson et al. 2025 AEJ: EP R&R; Humphries et al. 2024 JPE R&R)

My Household Finance Perspective On A Corporate Finance Topic

- **Corporations also** often have volatile incomes & limited liquid cash
 - Vulnerable to income/expenditure shocks



Note: Profit margin and cash buffer days measured from 2013 to 2017 in the cross-sectional sample.

Source:
JP Morgan Chase Institute (2019)

My Household Finance Perspective On A Corporate Finance Topic

- **Corporations also** often have volatile incomes & limited liquid cash
 - Vulnerable to income/expenditure shocks
- Motivates policies to provide temporary liquidity relief
 - Aim to **prevent illiquid-but-solvent business failures**
 - Potential **cost** of inefficiently **keeping insolvent firms in business** (zombies)
- This paper studies the trade-offs of one such policy:

Commercial rent forbearance during COVID-19

A Very Well-Polished Graduated JMP!

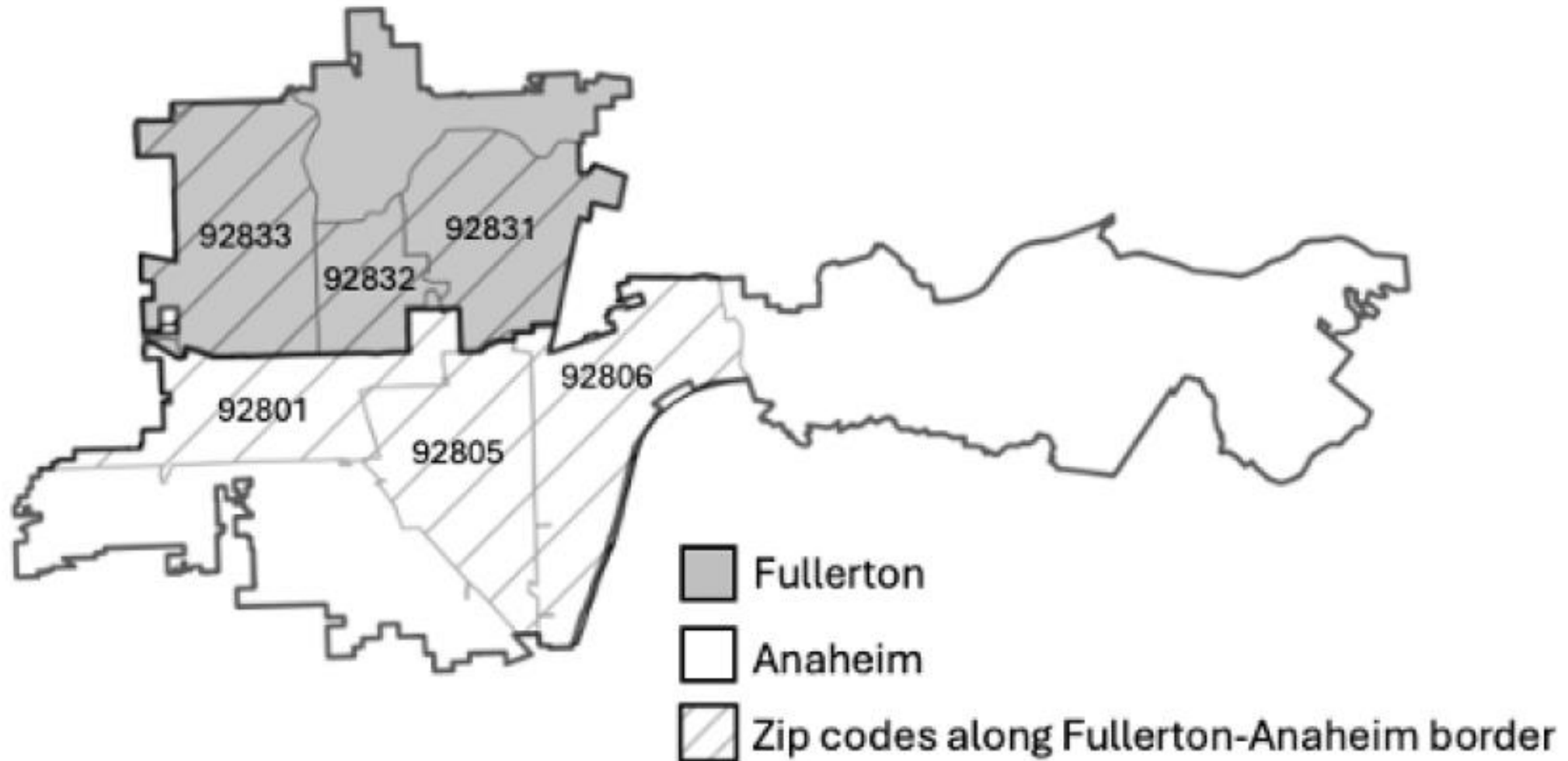
- Studies an important topic 
 - First-order classic literatures dating back to Schumpeter!
 - Learn about specific policy & COVID-19
- **Very** well-written 
- Neat data collection 
- I believe the paper's story! 
 - Moratorium  closure rate of low-liquidity businesses
 - No clear evidence of strategic zombie survival
 - Low reallocation suggests benefit of supporting low-liquidity businesses going
- **My Discussion:** Suggestions to make the results more convincing

#1: COVID-19

General Challenge Of COVID-19 Research: Heterogenous Impacts of Pandemic AND Many Simultaneous Policies Endogenous to Pandemic



Bordering Zip Codes Is A Nice Way To Address

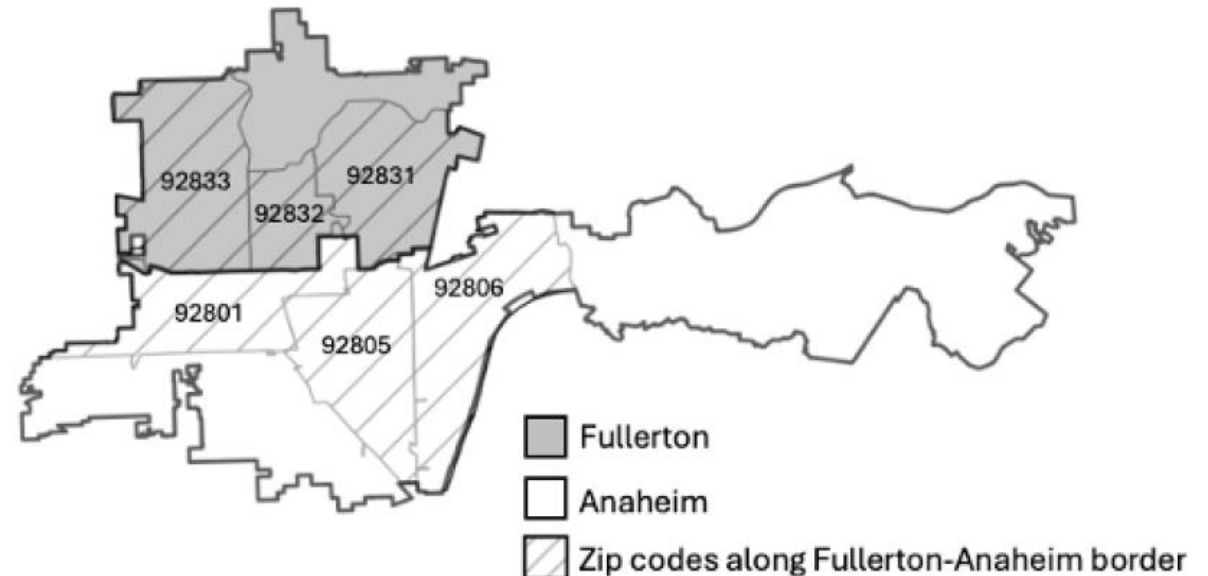


Do COVID-19 Impacts Vary Across Pairs?

- Would like to see COVID-19 impacts estimated over time
- I expect Anaheim's economy is more adversely affected (fewer tourists to Disney) than Fullerton ***irrespective of moratorium***

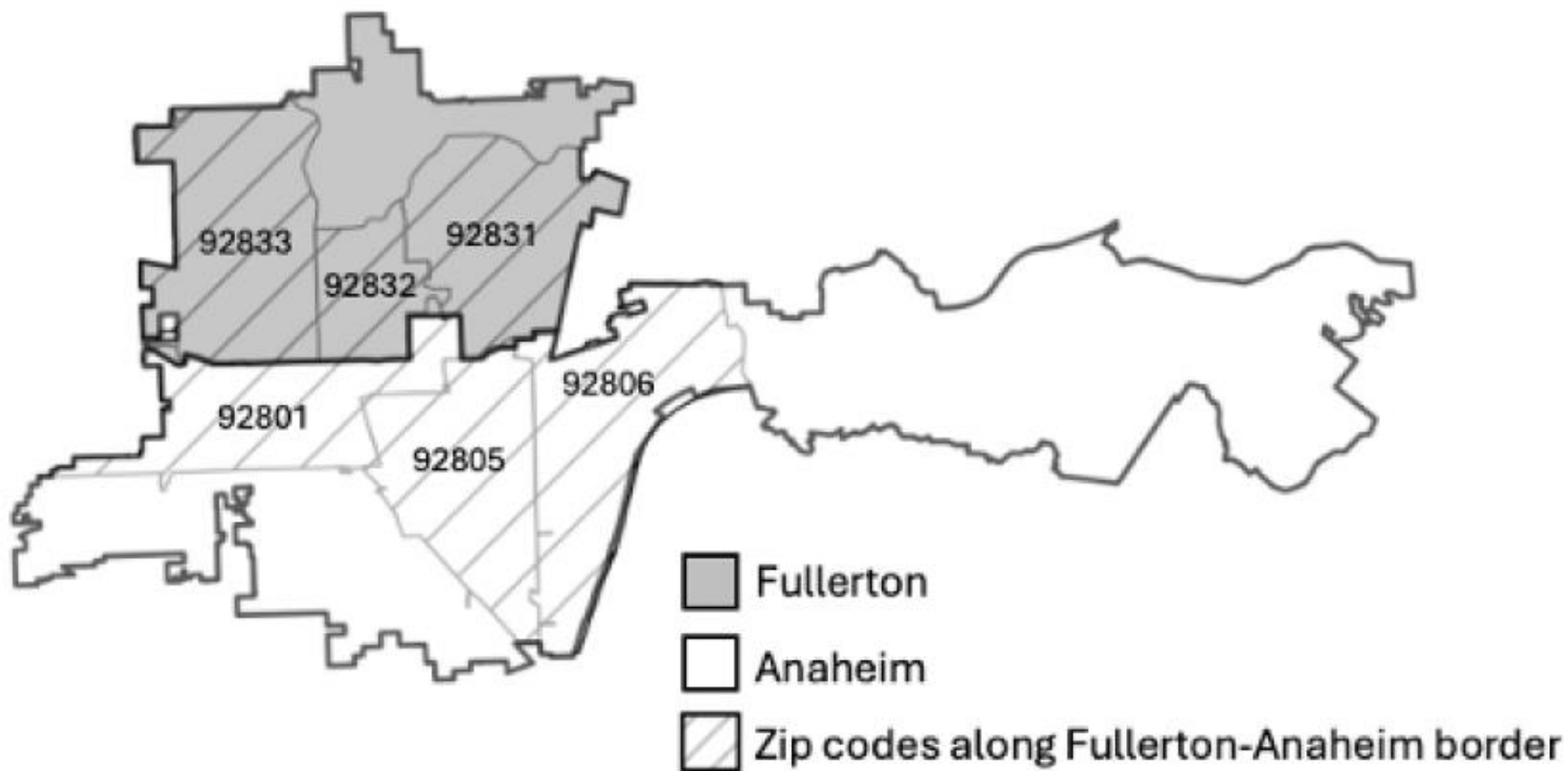
Suggestions?

- Examine COVID-19 outcomes
 - If no, effect. Reassuring!
 - If an effect, include time-varying COVID-19 and/or economics controls



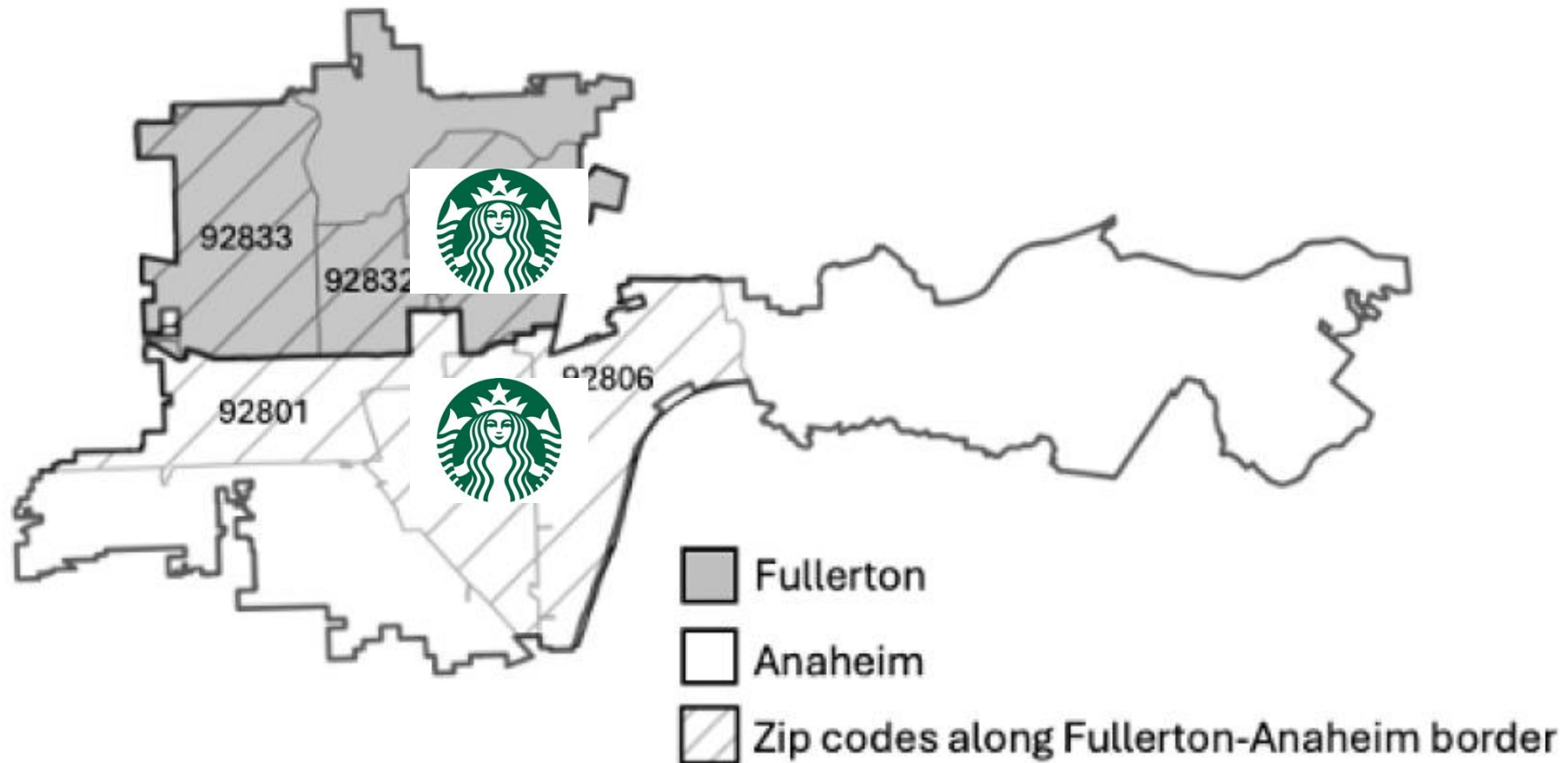
#2: Finer Comparison

Variation Compares Businesses In Same NAICS Sector Within City-City Pair e.g., “Accommodation & Food Services” Is 1 Group



Compare “Branded Businesses” e.g., Anaheim Starbucks vs. Fullerton Starbucks

- Applying same approach used for store openings
- Discuss how spillover effects may attenuate estimates here & existing designs



#3: Specifications

Business Closure Is Hard-to-Predict

	More liquidity-constrained sectors			Less liquidity-constrained sectors		
	Quarterly business closure (bps)			Quarterly business closure (bps)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated x Post	-8.729** (3.716)	-8.777** (3.719)	-8.803** (3.732)	2.110 (5.243)	2.208 (5.264)	2.269 (5.279)
Treated	Yes	Yes	Yes	Yes	Yes	Yes
Post	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects:						
City-city border	Yes	Yes	Yes	Yes	Yes	Yes
NAICS sector	No	Yes	Yes	No	Yes	Yes
Controls	No	No	Yes	No	No	Yes
Observations	2,586,512	2,586,512	2,586,512	835,148	835,148	835,148
R-squared	0.001	0.003	0.003	0.001	0.002	0.002

What Are The Baseline Business Closure Rates?

- Baselines Important To Interpret Magnitudes
- Expect Very Different Baselines Across Sectors

	More liquidity-constrained sectors			Less liquidity-constrained sectors		
	Quarterly business closure (bps)			Quarterly business closure (bps)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated x Post	-8.729** (3.716)	-8.777** (3.719)	-8.803** (3.732)	2.110 (5.243)	2.208 (5.264)	2.269 (5.279)
Treated	Yes	Yes	Yes	Yes	Yes	Yes
Post	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects:						
City-city border	Yes	Yes	Yes	Yes	Yes	Yes
NAICS sector	No	Yes	Yes	No	Yes	Yes
Controls	No	No	Yes	No	No	Yes
Observations	2,586,512	2,586,512	2,586,512	835,148	835,148	835,148
R-squared	0.001	0.003	0.003	0.001	0.002	0.002

Ideas To Tighten Analysis

Data Structure

- Is business no longer observed after closing? Suggest creating balanced panel
 - Binary only takes 1 in quarter of closing. Rare event! Closure is an absorbing state so in a balanced panel you could classify closures as 1s in quarters after closing
 - The less rare you make the binary outcome, the more power you have to detect

Ideas To Tighten Analysis

Data Structure

- Is business no longer observed after closing? Suggest creating balanced panel
 - Binary only takes 1 in quarter of closing. Rare event! Closure is an absorbing state so in a balanced panel you could classify closures as 1s in quarters after closing
 - The less rare you make the binary outcome, the more power you have to detect

Tighter Empirical Specifications

- **More granular time F.E.** instead of Post: Quarter or city-pair-by-quarter F.E.
- **More granular geographic F.E.** instead of additive city-city & sector: Industry-by-city-pair F.E. or create stacked DiD dataset (F.E. for each business)

Ideas To Tighten Analysis

Data Structure

- Is business no longer observed after closing? Suggest creating balanced panel
 - Binary only takes 1 in quarter of closing. Rare event! Closure is an absorbing state so in a balanced panel you could classify closures as 1s in quarters after closing
 - The less rare you make the binary outcome, the more power you have to detect

Tighter Empirical Specifications

- **More granular time F.E.** instead of Post: Quarter or city-pair-by-quarter F.E.
- **More granular geographic F.E.** instead of additive city-city & sector: Industry-by-city-pair F.E. or create stacked DiD dataset (F.E. for each business)

Scale up to get Treatment-on-the-Treated

- Any rough estimates of how many firms took up policy to delay rent?
- You are estimating ITT, but if low compliance then TOT may be much larger

Thank You!

 www.benedictgk.com

 benedictgk@rice.edu



JONES GRADUATE SCHOOL OF BUSINESS
VIRANI UNDERGRADUATE SCHOOL OF BUSINESS

Drafting Suggestions

- The abstract. The first two sentences are a great start. You then list a lot of results, but it isn't immediately clear to the reader the key takeaway of the paper. It would be helpful to include effect sizes for your main result, and the key takeaway/implication.
- It would be helpful to include in the introduction how widespread these moratorium were in the U.S. / or worldwide to help motivate the importance of the event you study.
- At some point in the paper, I would find it helpful if you were a bit more explicitly clear on the policy implication for how to respond to crises. Was this an effective policy? It would be worth briefly discussing other effects of the policy, especially negative effects that may outweigh the positive effects you find, that you are leaving for future research, e.g., the effects on commercial landlords (how do they cope with liquidity shortfall?), and post-moratorium rents.
- COVID-19 prompted a surge in restaurant productivity (Goolsbee, Syverson, Goldgof, & Tatarka, 2025). More granular than NAICS sector controls may help to control for this.
- I kept wondering, how many firms postponed rent. Do you have anything, even if anecdotal on the take0p rate? It may be worth a line noting that you are estimating the Intention-to-Treat. So, the effect on the subset of firms that take-up postponing rent may have a much larger Treatment-on-the-Treated effect.
- In Section 3.2, if you are first to validate this SafeGraph business openings and closing data, that seems more generally useful across economics and finance and something to emphasize. I would maybe add a line to the introduction as a methodological contribution of the paper. If others have used and validated the data, I would cite them as showing this is a robust data source.
- Last paragraph of section 6.2 was a lot to unpack. Break that paragraph up. Section 6 was the only section that didn't flow with the rest of the paper. It would help the reader to link it with the rest. Also, this was the only section that dragged, it felt as though you could make this section shorter with more of the institutional details in appendix to get the key point across, especially as that result is what we would expect given the economic incentives.
- Appendix doesn't seem to be referenced in the paper. I'm guessing this is JMP content put there temporarily!

Table Suggestions

- Table 1 has a lot of redundant information (e.g. Min, Med, and Max of binary variables not informative) so you could leave uninformative cells blank. A more streamlined table would just show mean and std. dev.. If you want more on the distribution of length this seems like an appendix figure.
- Table 2 would benefit from a total row that combines across sectors. I was also unclear whether this is all the data or just those used in the city pairs. More importantly, this table contains a lot of information for the reader to process. You just have a throwaway sentence in the paper to see summary statistics. If it is important, talk the reader through it. If it isn't important, summarize the key takeaway in the paper and move the table to an appendix.
- Regression Tables 3 to 8 need baseline means to interpret the estimates. I expect that the baseline means will be quite different for the heterogeneous groups in Tables 2 to 7.
- I really want to see a balance table that evaluates how similar the control and treatment businesses are in observable characteristics.
- I would like to see an appendix table that lists the 49 city-city border pairs.

Figure Suggestions

- Figure 1. I found the timeline an unnecessary distraction. The bottom figure is a much cleaner way to communicate the policy.
- Figure 2. The paper isn't using variation in the length of forbearance, so it may be cleaner to move this to an appendix and replace it with a simpler map just showing areas that were or were not affected. Currently the length is a distraction.
- Figure 3. Choose a different example. Anaheim immediately makes reader think of Disney. Also label which is treated and which is control.
- Figure 4. Why do the estimates go to zero quarters 8-12? Is this being driven by a subset of CEMs ending around this time?
- Figure 6. It is hard to visually compare panels A and B.
 - Putting them on one chart would enable the reader to more easily see what has changed from the short-to-long-term. E.g., have short-run and long-run in different shades or colors and then arrows showing where each industry point has moved from/to.

Typos

- Page 19 ocst
- Page 21 Equation has Business Opening (i,t) but paragraph below has Business Opening (i,q) and business b
- Equations have intercepts but I normally don't expect to see this with fixed effects
- A few times acronyms are not defined e.g., CEM in Figure 1, EIDL not defined in 6.1 and Figure 7
- Appendix page numbers repeat